

Lab 1 - Knowledge Is Power (KIP)

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CS 411

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Submission Date: 9/29/2026

Version 1

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1. Introduction

Knowledge Is Power (KIP) is a platform that helps young adults learn about laws they may face in everyday life. This section explains the problem KIP is trying to solve, what the solution should include, and how the product can help users make better decisions.

1.1. Societal Problem

Young people between the ages of 16 and 19 increasingly encounter laws involving driving, digital conduct, employment, school, ownership, contracts, and personal safety without fully understanding how these laws work or the consequences that may result from violating them. As teenagers gain more independence and transition into adulthood, this lack of legal awareness can lead to poor decisions and serious consequences.

Despite this need, law-related education remains largely absent from K-12 curricula, leaving many young adults underprepared for real-world legal situations. Teen drivers face higher safety risks that are often connected to poor judgment and decision-making [4.1, 4.2, 4.3, 4.4].

1.2. Needed Solution Characteristics

A solution to this problem should teach legal concepts through interactive activities that require users to apply laws to realistic situations rather than simply memorize definitions. To remain accessible to young users, the platform should use clear and consistent language, concise explanations, mobile-friendly web design, and low-stakes practice games that students can complete independently.

The solution should also explain the connection between each lesson and its official legal sources. Legal vocabulary should be introduced in context so that users can recognize practical legal terms that may appear across different situations and topics.

1.3. Product Introduction

Knowledge Is Power (KIP) is a digital legal literacy platform designed to teach essential everyday laws to young adults. It is intended to serve as a supplemental educational resource for students, parents, teachers, and school administrators.

KIP's goal is to help young adults recognize legally significant situations and understand the possible consequences of their decisions so they can make more informed choices. The prototype will demonstrate these goals through a web-based service that supports user accounts, learning modules, interactive scenarios, progress tracking, and personalized recommendations based on user performance.

2. KIP (Knowledge Is Power) Product Description

KIP is structured as a web-based educational service built around learning modules, interactive games, user progress data, and source-backed legal content. The system presents users with short misconception-based quizzes and longer branching scenarios, then uses user performance to recommend future practice.

KIP's software components support account access, scenario delivery, feedback generation, content curation, source referencing, and persistent progress tracking. Together, these features are intended to assess users' current legal understanding, strengthen their ability to recognize legally significant situations, and reinforce informed decision-making through repeated low-stakes practice.

2.1. Key Product Features and Capabilities

This section describes KIP's main features, goals, games, and learning capabilities.

2.1.1. Solution Overview

KIP uses focused games, realistic examples, curated media, and official legal resources to educate young adults about practical legal boundaries and everyday legal situations.

2.1.2. Goals and Objectives

KIP aims to assess, improve, and reinforce legal knowledge among technology-friendly young adults through repeatable, low-stakes interactive activities. Its user experience (UX) is designed to improve recognition of legal situations, strengthen decision-making skills, and increase long-term retention of legal concepts.

2.1.3. Interactive Gamification

KIP uses focused, low-stakes games to improve users' confidence when making decisions involving legal situations. These games allow users to apply knowledge in realistic scenarios without facing real-world consequences.

2.1.3.1. "Fact or Fiction?" Game

"Fact or Fiction?" is a short-form game that presents users with common legal beliefs and misconceptions. Users may answer "true," "false," or "context-dependent."

This approach helps users challenge assumptions, recognize legal nuance, and develop foundational knowledge that prepares them for longer and more detailed scenarios.

2.1.3.2. "Choose the Outcome" Game

"Choose the Outcome" is a longer multiple-choice game that allows users to interact with people, objects, and documents in order to learn details about a scenario before making a decision.

Each choice demonstrates possible consequences and explains its legal significance. This encourages users to slow down, evaluate the available context, and understand what information may affect a legal outcome.

2.1.4. Progress Tracking and Analytics

KIP tracks users' topical strengths, weaknesses, and completion progress. This information allows the system to recommend relevant games and modules based on previous performance.

Progress information also allows future game sessions to present questions, scenarios, and legal concepts that meaningfully challenge the user in areas where additional practice may be needed.

2.2. Major Components

KIP consists of presentation, application, data, and hardware layers that work together to provide the web-based learning experience.

2.2.1. Presentation Layer

The presentation layer provides the user-facing parts of KIP, including account access, browsing, and gameplay.

2.2.1.1. Technology Stack

The presentation layer will use Angular, TypeScript, JavaScript, Bootstrap, and HTML.

2.2.1.2. Account Portal

The Account Portal allows existing users to log in and new users to create an account. Users may also access certain games without an account in order to experience KIP with limited functionality.

2.2.1.3. Module Browser

The Module Browser displays available legal learning modules, including topics such as traffic law, online conduct, school and work, and personal safety. Users can select a topic and view their progress within each module.

2.2.1.4. Game Browser

The Game Browser displays available games and prioritizes recommendations based on the user's progress, performance history, and demonstrated strengths or weaknesses in the legal topics covered by each game.

2.2.1.5. Scenario Player

The Scenario Player is the primary interface users interact with while playing KIP's games. It presents interactive questions, branching scenarios, feedback, and source references during gameplay.

2.2.2. Application Layer

The application layer manages KIP's main logic, authentication, scenarios, feedback, and progress tracking.

2.2.2.1. Technology Stack

The application layer will use Java, Spring Boot, and JHipster.

2.2.2.2. Authentication System

The Authentication System manages user login, account creation, guest access, and session validation. It determines whether a user is accessing KIP through a registered account or as a limited guest and controls which features are available based on that status.

2.2.2.3. Scenario Engine

The Scenario Engine loads and manages the interactive content used during KIP's games. During a play session, the engine assembles a match containing multiple rounds.

Each round establishes unique choices, scenario branches, consequences, and game states that appear in sequence. The content presented is primarily informed by the selected game and the user's previous actions.

2.2.2.4. Feedback System

The Feedback System provides explanations after users make decisions. It identifies whether an answer is correct, incorrect, or context-dependent and explains the legal reasoning behind the result.

The system helps users understand why a particular choice may be safe, risky, incorrect, or incomplete.

2.2.2.5. Curation System

The Curation System organizes and recommends learning content based on the user's progress and performance history. It determines which games, modules, and scenarios should be prioritized based on areas where the user needs additional practice.

Users will still be able to freely browse and select any available content.

2.2.2.6. Source Reference System

The Source Reference System connects game content and feedback to official legal sources, educational resources, and relevant legal vocabulary. It allows users to see where KIP's information comes from and, when desired, review the original legal language and documents behind a lesson.

The system also supports content maintenance by tracking changes to legislation, updating references when necessary, and flagging KIP content that may be outdated. Administrators can then review, revise, or disable outdated content.

2.2.3. Data Layer

The data layer stores and manages the information needed by the KIP system.

2.2.3.1. Technology Stack

The data layer will use MySQL.

2.2.3.2. Databases

KIP will include the following databases:

- User Database
- Progress Database
- Module Database
- Game Database
- Legal Content Database

2.2.3.3. Functions

The data layer stores user account information, user progress records, learning modules and associated content, game scenarios and gameplay information, and legal and educational resources.

2.2.4. Hardware Layer

The hardware layer includes the server and user devices required to operate and access KIP.

2.2.4.1. Server-Side Components

A virtual private server will host the Java application, manage server-side processing, and provide access to the MySQL database and other application resources.

2.2.4.2. Client-Side Components

Users will access KIP using consumer devices equipped with a modern web browser and an internet connection.

2.2.4.3. Functions

The hardware components support the execution of Java application code, storage of persistent data, processing of user requests, and delivery of the web-based interface to users.

3. Glossary

CDC: Centers for Disease Control and Prevention

KIP: Knowledge Is Power

MFCD: Major Functional Component Diagram

UX: User Experience

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